

Convergence of the DFP Algorithm

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April 7, 2026

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The DFP Algorithm

The DFP Algorithm is the first quasi-Newton method for unconstrained optimization

$$\min_{x \in \mathbb{R}^n} f(x). \quad (1.1)$$

The algorithm is as follows.

The DFP Algorithm

Step 0. Given $x_1 \in \mathbb{R}^n$; $B_1 \in \mathbb{R}^{n \times n}$ positive definite; $k := 1$.

Step 1. Compute $g_k = \nabla f(x_k)$;

if $g_k = 0$ then stop;

set $d_k = -B_k^{-1}g_k$;

Step 2. Carry out a line search along d_k , getting $\alpha_k > 0$;

set $x_{k+1} = x_k + \alpha_k d_k$;

Step 3. Set

$$B_{k+1} = B_k - \frac{B_k s_k y_k^T + y_k s_k^T B_k}{s_k^T y_k} + \left(1 + \frac{s_k^T B_k s_k}{s_k^T y_k}\right) \frac{y_k y_k^T}{s_k^T y_k}, \quad (1.2)$$

where

$$s_k = \alpha_k d_k \quad (1.3)$$

$$y_k = g_{k+1} - g_k \quad (1.4)$$

Step 4. $k := k+1$; go to Step 1.

The DFP Algorithm

If we replace(1.2)with

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{s_k^T y_k}$$

then we get the BFGS algorithm.

The Wolfe Line Search

The line search in Step 2 requires step lengths d_k to satisfy certain line search conditions.

If exact line search is used, α_k satisfies

$$f(x_k + \alpha_k d_k) = \min f(x_k + \alpha d_k). \quad (2.1)$$

For inexact line search, the Wolfe conditions usually serves as an alternative for.

The Wolfe Line Search consists of two conditions, the Armijo Condition that ensures sufficient decrease and the Curvature Condition ensures the steps size to be "good" enough.

Armijo Condition

The Armijo condition is a sufficient decrease condition that ensures that the step size chosen in the line search leads to a sufficient decrease in the objective function. The formula for the Armijo condition is given by:

$$f(x_k + \alpha d_k) \leq f(x_k) + c_1 \alpha \nabla f(x_k)^T d_k, \quad (2.2)$$

where f is the objective function, x_k is the current point, d_k is the search direction, α is the step size, and c_1 is a constant in the range $(0, 1)$. The Armijo condition ensures that the step size chosen in the line search leads to a sufficient decrease in the objective function, which is important for the convergence of optimization algorithms.

Curvature Condition

The curvature condition is a condition that ensures that the step size chosen in the line search is not too small. It is given by the formula:

$$\nabla f(x_k + \alpha d_k)^T d_k \geq c_2 \nabla f(x_k)^T d_k, \quad (2.3)$$

where c_2 is a constant in the range $(c_1, 1)$. The curvature condition ensures that the step size chosen in the line search is not too small, which can lead to slow convergence or even divergence of optimization algorithms.

- Powell[3] showed that if $f(x)$ is a convex function and if line searches are exact in all iterations, then the DFP algorithm will either stop at a global minimum or generate a sequence of iterates that converges to a global minimum.
- For inexact line search, Powell[4] showed that the BFGS method is globally convergent.
- Byrd, Nocedal and Yuan[5] showed the global convergence of all the methods in the convex Broden family except the DFP method.

- In the 1990s, Yuan[6](improved by Yin and Han[7] three years later), Xu and Liu[8] gave sufficient conditions for the global convergence of the DFP method with objective function being uniformly convex respectively.
- In 2002, Liu, Han and Xu[9] proved that convergence of DFP algorithm with Wolfe line search holds for quadratic uniformly convex functions.
- In 2024, Luo, Wu and He[10] derived a new DFP correction formula and proposed a new class of DFP algorithm under the strong Wolfe condition, improving the problem of linesearch failure due to accuracy and proving global convergence of the improved algorithm under certain assumptions.

Some sufficient conditions for the global convergence of the DFP algorithm

In this section, we will talk about some sufficient conditions for the global convergence of the DFP Algorithm given by Yuan[6], Xu and Liu[8], Yin and Han[7].

All these works were established under the following hypothesis (H):

- ① $f(x)$ is twice continuously differentiable and uniformly convex.

Namely, there exists a constant $c_3 \geq 0$ such that

$$d^T \nabla^2 f(x) d \geq c_3 \|d\|_2^2, \quad \forall d, x \in \mathbb{R}^n. \quad (3.1)$$

- ② The Wolfe Conditions are satisfied in all iterations.

Using the well-known Trace Relation

$$\text{tr}(B_{k+1}) = \text{tr}(B_k) - 2 \frac{s_k^T B_k y_k}{s_k^T y_k} + \left(1 + \frac{s_k^T B_k s_k}{s_k^T y_k}\right) \frac{\|y_k\|_2^2}{s_k^T y_k}, \quad (3.2)$$

Yuan obtained the following result:

Theorem (Yuan, 2.1)

If $\{\alpha_k\}$ are bounded away from zero, and if $\text{Tr}(B_k)$ are monotonically increasing, the DFP method converges to the unique solution.

Yuan's work in the 1990

Subsequently, Yuan got the following:

Theorem (Yuan,2.2)

Assume that the hypothesis (H) holds. If the sequence x_k generated by the DFP algorithm satisfies

$$\|g_{k+1}\|_2 \leq \|g_k\|_2 \quad (2.21)$$

for all large k , then x_k converges to the unique minimum of $f(x)$.

Yuan's work in the 1990

In addition, we have

Theorem (Yuan,2.3)

If

$$\sum_{k=1}^{\infty} \|s_k\| < \infty. \quad (2.43)$$

then the DFP algorithm converges to the solution.

Theorem (Yuan,2.4)

Assume that the conditions in Hypothesis (H) hold. Let the sequence $\{x_k, k = 1, 2, \dots\}$ be generated by the DFP method. If

$$y_k^T B_k g_k \leq 0 \quad (2.47)$$

for all k , then $\{x_k\}$ either terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

In this essay, Xu and Liu established the following sufficient conditions for the global convergence of the DFP Algorithm.

Theorem (Xu and Liu,2.3)

Assume the conditions in Hypothesis (H) hold. If

$$y_k^T B_k^2 g_k \leq 0 \quad (2.10)$$

holds for all large k , then the sequence $\{x_k, k = 1, 2, \dots\}$ generated by the DFP method either terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

Theorem (Xu and Liu,2.4)

The condition (2.10) in Theorem (Xu and Liu,2.3) can be replaced by

$$\sum_{i \in I_k} (y_i^T B_i^2 g_i)^2 \leq C_2 \sum_{i=1}^k \frac{1}{\|d_i\|_2^3},$$

where $I_k = \{i : y_i^T B_i^2 g_i > 0, i < k\}$, $C_2 > 0$.

Yin and Han's Improvement

In this essay, Yin and Han established the following theorem:

Theorem (Yin and Han,2.3)

Assume that the conditions in Hypothesis (H) hold. Let the sequence $\{x_k, k = 1, 2, \dots\}$ be generated by the DFP method. If

$$\text{tr}(B_k) \leq c_6 \sqrt{\sum_{i=1}^k \frac{1}{\|d_i\|^2}}, \quad \forall k \geq 1, \quad (2.14)$$

where $c_6 > 0$ constant, then $\{x_k\}$ terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

Yin and Han's Improvement

Then we get the following theorem:

Theorem (Yin and Han,2.4)

Under the hypothesis in theorem (Yin and Han,2.3), if there exists a constant $c_7 > 0$ such that

$$\text{Tr}(B_k) \leq c_7 k, \quad \forall k \geq 1,$$

then $\{x_k\}$ terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

From the theorems above, we can get some sufficient conditions for the global convergence of the DFP Algorithm.

Yin and Han's Improvement

Theorem (Yin and Han,3.1)

Assume that the conditions in Hypothesis (H) hold. Let the sequence $\{x_k, k = 1, 2, \dots\}$ be generated by the DFP method. If

$$\sum_{i \in I_k} \frac{g_i^T y_i}{d_i^T y_i} \leq c \sqrt{\sum_{i=1}^k \frac{1}{\|d_i\|^2}}, \quad \forall k \geq 1,$$

where $c > 0$ constant, $I_k = \{i \mid 1 \leq i \leq k, g_i^T y_i > 0\}$, then $\{x_k\}$ terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

Yin and Han's Improvement

Theorem (Yin and Han,3.2)

Assume that the conditions in Hypothesis (H) hold. Let the sequence $\{x_k, k = 1, 2, \dots\}$ be generated by the DFP method. If

$$g_k^T y_k \leq 0, \quad \forall k \geq 1,$$

then $\{x_k\}$ terminates at the unique minimum x^ of $f(x)$ or converges to x^* .*

Note that

$$\begin{aligned} g_k^T y_k &= g_k^T (g_{k+1} - g_k) = g_k^T g_{k+1} - \|g_k\|^2 \leq \frac{1}{2} \left(\|g_k\|^2 + \|g_{k+1}\|^2 \right) - \|g_k\|^2 \\ &= \frac{1}{2} \left(\|g_{k+1}\|^2 - \|g_k\|^2 \right) \leq 0, \quad \forall k \geq 1 \end{aligned}$$

then from theorem (Yin and Han,3.2),we obtain theorem (Yuan,2.2), indicating that the conditions in theorem (Yuan,2.2) are stronger than those in theorem(Yin and Han,3.1).

Yin and Han's Improvement

Theorem (Yin and Han,3.4)

Assume that the conditions in Hypothesis (H) hold. Let the sequence $\{x_k, k = 1, 2, \dots\}$ be generated by the DFP method. If

$$\sum_{k \in J} g_k^T y_k < +\infty, \quad (3.10)$$

where $J = \{k \mid g_k^T y_k > 0, k = 1, 2, \dots\}$, then $\{x_k\}$ terminates at the unique minimum x^ of $f(x)$.*

If we replace (3.10) with

$$\sum_{k=1}^{\infty} |g_k^T y_k| < +\infty, \quad (3.12)$$

theorem (Yin and Han,3.4) also holds.

Yin and Han's Improvement

Note that

$$\left| g_k^T y_k \right| \leq \|g_k\| \|y_k\| \leq \eta \|y_k\| \leq \eta c_4 \|s_k\| ,$$

then from theorem (Yin and Han,3.4),we obtain theorem (Yuan,2.3), indicating that the conditions in theorem (Yuan,2.3) are stronger than (3.12).

Additionally, we can also obtain (Yuan,2.1) by (Yin and Han,3.4).

Theorem

Assume that the conditions in Hypothesis (H) hold. If there exists a constant $b > 0$ such that

$$\|g_k\| \leq b \|d_k\|, \quad \forall k \geq 1,$$

then $\{x_k\}$ terminates after finite steps or converges to the unique minimum x^ of $f(x)$.*

Yin and Han's Improvement

Under the hypothesis (H), Yuan[6] gave an sufficient condition

$$y_k^T B_k g_k \leq 0, \quad \forall k \geq 1.$$

Xu and Liu[8] suggested that

$$y_k^T B_k^2 g_k \leq 0, \quad \forall k \geq 1.$$

Yin and Han[7] showed that

$$y_k^T B_k^0 g_k \leq 0, \quad \forall k \geq 1.$$

One may propose that if there exists an integer $m \geq 0$ such that

$$y_k^T B_k^m g_k \leq 0, \quad \forall k \geq 1$$

then $\{x_k\}$ terminates after finite steps or converges to the unique minimum x^* of $f(x)$. However, whether this proposition holds is still unknown.

In this talk, we discover the origin and recent progress in the global convergence of the DFP algorithm.

Then we review some essays giving sufficient conditions of convergence under some hypothesis. ((i) the functions are twice continuously differentiable and uniformly convex (ii) Wolfe conditions are satisfied in all iterations) Finally, we make some comparisons of the results, and point out an open problem to be solved.

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Thanks For Listening